10.21 The One-Loop φ^3 $2 \to 2$ Amplitude at High Energy

LecNote14 pp. 6–8 (Srednicki §20) assemble the one-loop $2 \to 2$ amplitude of φ^3 theory in $d = 6$ from LecNote12–13's self-energy, vertex and box, and on p. 8 quote its limit at high energy and fixed angle. Page numbers without a LecNote refer to LecNote14. Students ask:

- "Where do p. 8's $3 - \pi\sqrt{3}$, -3 and $\pi^2 + \ln^2(s/t)$ come from, and what is c? The boxed line says $6\pi^2 + \pi\sqrt{3} - 39 = 2.33$; I get 25.7."
- "Which m^2 may I drop at $s \gg m^2$? Without m^2, what divides s in $\ln(-s/m^2)$?"
- "Where does the $i\epsilon$ go, and why is T complex now?"
- "'If α is large this correction is bigger than the tree.' So a small α is safe?"

In short: m^2 drops out where it competes with s, t, u and stays where the renormalization conditions put it; three Feynman-parameter integrals then give p. 8, with $c = (6\pi^2 + \pi\sqrt{3} - 39)/11$; and no α is safe from $\frac{11}{12}\alpha \ln(s/m^2)$.

Conventions and $\mathcal{L}$ are those of §10.14 (metric $(-,+,+,+)$, $\tilde{\Delta} = 1/(k^2 + m^2 - i\epsilon)$, $\alpha = g^2/(4\pi)^3$), with $C = Z_g - 1$ fixed by $V_3(0,0,0) = g$ and A, B by $\Pi(-m^2) = \Pi'(-m^2) = 0$ (LecNote12–13); $s = -(k_1 + k_2)^2$, $t = -(k_1 - k_1')^2$, $u = -(k_1 - k_2')^2$; $\int dF_n \equiv (n-1)! \int d^n x\, \delta(1 - \sum_i x_i) = 1$.

The pieces

The skeleton expansion (LecNote14 pp. 5–6; §10.20) writes the amplitude as a tree of the exact propagator $\tilde{\mathbf{\Delta}}(-s) = 1/[m^2 - s - \Pi(-s)]$, the exact vertex V_3 and the 1PI four-point function V_4. With $\frac{1}{i}(iV_3)^2 = iV_3^2$ (p. 6),

$$T = \sum_{x=s,t,u} V_3(x)^2\, \tilde{\mathbf{\Delta}}(-x) + V_4(s,t,u), \tag{10.21.1}$$

where $V_3(x)$ has two legs on shell and the third at $k^2 = -x$; external legs give exactly 1 (§10.11). At tree level this is (10.11.11); with each piece at one loop it is T to $O(g^4)$. The pieces are p. 7's, with its slips corrected ($-s$ for $-s^2$, the α in V_3, the sign in D_0). They are Srednicki's (20.4)–(20.6), in the same symbols: $\Pi(-s)$ with $\ln(D_2/D_0)$, $V_3(s)/g = 1 - \frac{\alpha}{2}\int dF_3 \ln(D_3/m^2)$, and V_4 with the prefactor $g^2\alpha/6$, where, as in his (20.8), (20.10) and (20.11),

$$D_2 = m^2 - x(1-x)s, \qquad D_3 = [1 - (x_1 + x_2)x_3]m^2 - x_1x_2s,$$
$$D_4(s,t) = [1 - (x_1 + x_2)(x_3 + x_4)]m^2 - x_1x_2s - x_3x_4t.$$

In LecNote12's on-shell Π, $D_0 = [1-x(1-x)]m^2$ is D_2 at $s=m^2$, which makes $\Pi(-m^2) = \Pi'(-m^2) = 0$. D_3 is LecNote13's $D = x_1x_3k_1^2 + x_2x_3k_2^2 + x_1x_2k_3^2 + m^2$ at $k_1^2 = k_2^2 = -m^2$, $k_3^2 = -s$; the m^2 under its logarithm encodes $V_3(0,0,0) = g$, since $D = m^2$ at zero momenta. In V_4, $(ig)^4(1/i)^4 = g^4$, and $\frac{1}{6} = \Gamma(1)/\Gamma(4)$ comes from (10.18.1): it is not a symmetry factor.

For D_4, give the box's lines momenta $l + p_i$ in cyclic order and parameters y_i: $D = m^2 + \sum_{i<j} y_iy_j(p_i - p_j)^2$. Neighbours differ by one external momentum and give $-m^2(y_1+y_3)(y_2+y_4)$, opposite lines by two, and $(y_1, y_3, y_2, y_4) \to (x_1, \dots, x_4)$ gives D_4. The cyclic orders 1234, 1324, 1243 ($k_3 = -k_1'$, $k_4 = -k_2'$) carry (s,u), (t,u), (s,t): as $D_4(s,t) = D_4(t,s)$, these are all three pairs, and T is symmetric under every permutation of s,t,u. p. 7's "cyclic, not exchange" says how the terms are generated, not that a symmetry is missing.

High energy at fixed angle

In the c.m. frame $t, u = -\frac{1}{2}(s-4m^2)(1 \mp \cos\theta)$: at fixed $\theta \neq 0, \pi$, $|t|, |u| \gg m^2$ too (p. 7's "$\geq$" means "$\gg$"), and s/m^2 is the only large ratio. Drop m^2 where it competes with s,t,u (in D_2, D_3, D_4, $m^2 - s$, $s+t+u$), at a cost $O(m^2/s)$ up to logarithms. Keep it where nothing competes: D_0 and the m^2 in $\ln(D_3/m^2)$ come from the renormalization conditions and set the scale of every logarithm.

Self-energy. $D_2 \to -x(1-x)s$. With $\int_0^1 dx\, x(1-x) = \frac{1}{6}$, $\int_0^1 dx\, x(1-x)\ln[x(1-x)] = -\frac{5}{18}$ and, from D_0, $\int_0^1 dx\, x(1-x)\ln(1-x+x^2) = \frac{\pi\sqrt{3}}{6} - \frac{17}{18}$,

$$\Pi(-s) \to -\frac{\alpha s}{12}\Big[\ln\frac{-s}{m^2} + 3 - \pi\sqrt{3}\Big], \tag{10.21.2}$$

which is (10.14.6) at $k^2 = -s$; to $O(\alpha)$, $\tilde{\mathbf{\Delta}}(-s) \to -\frac{1}{s}[1 - \Pi(-s)/s]$.

Vertex. $D_3 \to -x_1x_2s$, and $\int dF_3 \ln(x_1x_2) = 2\int dF_3 \ln x_1 = 4\int_0^1 dx\,(1-x)\ln x = -3$ (the $O(1)$ LecNote13 p. 4 drops), so $V_3(s)/g \to 1 - \frac{1}{2}\alpha[\ln(-s/m^2) - 3]$.

Box. $D_4(s,t) \to x_1x_2S + x_3x_4T$ with $S = -s$, $T = -t$. For $S, T > 0$,

$$\int dF_4 \frac{1}{x_1x_2S + x_3x_4T} = \frac{3}{S+T}\Big[\pi^2 + \ln^2\frac{S}{T}\Big]. \tag{10.21.3}$$

To see it, put $(x_1,x_2,x_3,x_4) = \big(y,\, 1-y,\, \lambda z,\, \lambda(1-z)\big)/(1+\lambda)$; the left side becomes $6\int_0^1 dy \int_0^1 dz \int_0^\infty d\lambda\, \lambda(1+\lambda)^{-2}/[y(1-y)S + \lambda^2 z(1-z)T]$. Both sides are T^{-1} times a function of $\rho = S/T$; integrate them against $\rho^{\sigma-1}d\rho$, $0 < \sigma < 1$, with $f(\sigma) \equiv \int_0^\infty d\rho\, \rho^{\sigma-1}/(1+\rho) = \pi/\sin\pi\sigma$. On the left, $\int_0^\infty d\rho\, \rho^{\sigma-1}/(\rho a + b) = f\, a^{-\sigma}b^{\sigma-1}$ leaves $B(1-\sigma,1-\sigma)B(\sigma,\sigma)B(2\sigma,2-2\sigma) = f^2$,

so $6f^3$ in all. On the right the $\ln^2\rho$ brings f'', and $3(\pi^2 f + f'') = 6f^3$ again: equal Mellin transforms, equal functions.

The $i\epsilon$. The propagators' $-i\epsilon$'s add up to $D - i\epsilon$, and every invariant enters D_2, D_3 and D_4 with a non-positive coefficient, so the rule is $s, t, u \to s, t, u + i\epsilon$ (§10.16; p. 8's red note). Both sides of (10.21.3) are analytic in S off the negative real axis, so it holds at $S = -s - i\epsilon$ too. For $s > 4m^2$, $t, u < 0$: $\ln(-s/m^2) = \ln(s/m^2) - i\pi$, $\ln(-t/m^2)$ and $\ln(-u/m^2)$ are real, and ratios are differences, $\ln(s/t) = \ln(s/|t|) - i\pi$, $\ln(u/s) = \ln(|u|/s) + i\pi$. T is complex only through the pieces that contain s, the one channel whose intermediate particles can be on shell. And $\operatorname{Im} T > 0$: at this order unitarity gives $\operatorname{Im} T = \frac{1}{2}\cdot\frac{1}{2}\int d\Phi_2\, T_0(k_1k_2 \to q_1q_2)\, T_0(q_1q_2 \to k_1'k_2')$ (the second $\frac{1}{2}$ for identical particles), and the tree (10.11.11) is positive for all $s > 4m^2$ and angles.

Assembly, and the constant c

To $O(\alpha)$ these limits make the s-channel term

$$V_3(s)^2\,\tilde{\mathbf{\Delta}}(-s) \to -\frac{g^2}{s}\Big\{1 - \frac{11}{12}\,\alpha \ln\frac{-s}{m^2} + \alpha\Big[3 + \frac{3-\pi\sqrt{3}}{12}\Big]\Big\}: \tag{10.21.4}$$

$\frac{11}{12} = 2\cdot\frac{1}{2} - \frac{1}{12}$, from two vertices and a propagator, whose logarithms have opposite signs. Of the boxes, (10.21.3) puts $1/s$ only into the (t,u) one: with $-(t+u) = s - 4m^2 \to s$, it is $\frac{g^2\alpha}{2s}[\pi^2 + \ln^2(t/u)]$. Collecting the $1/s$ terms, and the others by $s \to t \to u \to s$,

$$\begin{aligned} T &= g^2\big[F(s,t,u) + F(t,u,s) + F(u,s,t)\big], \\ F(s,t,u) &= -\frac{1}{s}\Big\{1 - \frac{11}{12}\,\alpha\Big[\ln\frac{-s}{m^2} + c\Big] - \frac{\alpha}{2}\ln^2\frac{t}{u}\Big\}, \\ c &= -\frac{12}{11}\Big[3 + \frac{3-\pi\sqrt{3}}{12} - \frac{\pi^2}{2}\Big] = \frac{6\pi^2 + \pi\sqrt{3} - 39}{11} = 2.3326, \end{aligned} \tag{10.21.5}$$

up to $O(g^6)$ and $O(m^2/s)$. p. 8's boxed 2.33 is this c, without its $/11$ ($6\pi^2 + \pi\sqrt{3} - 39 = 25.66$). The table compares (10.21.5) with the exact amplitude (masses kept, Srednicki's (20.4)–(20.6) integrated numerically), $T = T_0 + \alpha T_1 + O(\alpha^2)$: $\operatorname{Re} sT_1$ falls by $\frac{11}{12}sT_0 = 3.97$ per unit of $\ln s$ (exact: 3.965 from 10^5 to 10^6); $\operatorname{Im} sT_1$ is the unitarity integral. The last column is the α at which the one-loop term equals the tree.

s/m^2	sT_0/g^2	(10.21.5)	sT_1/g^2	(10.21.5)	$\lvert T_0/T_1\rvert$
10^2	4.360	4.556	$0.53 + 5.64i$	$3.28 + 7.65i$	0.77
10^4	4.334	4.335	$-15.06 + 6.52i$	$-14.95 + 6.55i$	0.26
10^5	4.333	4.334	$-24.11 + 6.54i$	$-24.09 + 6.54i$	0.17
10^6	4.333	4.333	$-33.24 + 6.54i$	$-33.23 + 6.54i$	0.13

One-loop φ^3 $2 \to 2$ at $\cos\theta = \frac{1}{2}$: exact, then asymptotic

The large logarithm. At fixed angle c and $\ln^2(t/u)$ are numbers, but $\frac{11}{12}\alpha\ln(s/m^2)$ reaches 1 at $\sqrt{s} = m\,e^{6/(11\alpha)}$ ($6.2\,m$, $234\,m$, $5.5\times10^4\,m$ for $\alpha = 0.3$, 0.1, 0.05), while the constant, $\frac{11}{12}c\,\alpha = 2.14\,\alpha$, does so only at $\alpha \approx 0.47$ (p. 8's "if α is large"). The logarithm defeats any α. Its m^2 comes from the renormalization conditions; trading it for $\mu \sim \sqrt{s}$ and running the coupling is the subject of §10.24, which accounts for the $\frac{11}{12}$.